

DATA INTERPRETATION MASTER GUIDE

Comprehensive Concepts, 12 Solved Examples & 25 Practice Questions

1. Data Interpretation Theory & Concepts

What is Data Interpretation (DI)?

Data Interpretation refers to the process of making sense out of a collection of data that has been processed. This collection may be represented in various graphical formats like Tables, Bar Graphs, Pie Charts, Line Graphs, and Caselets.

Important Calculation Tools

- **Percentages:** Essential for finding increase/decrease, ratios, and shares.
$$\text{Percentage} = (\text{Value} / \text{Total Value}) \times 100$$
$$\% \text{ Change} = ((\text{New Value} - \text{Old Value}) / \text{Old Value}) \times 100$$
- **Ratios & Proportions:** Comparing quantities of the same units. Simplification of data streams.
- **Averages:** Sum of observations divided by the total number of observations.
- **Growth Rate / CAGR:** Used across consecutive years to find compounding annual growth.

Common Types of DI Graphs

- **Tables:** Data organized neatly in rows and columns. High accuracy required in tracking cross-sections.
- **Bar Graphs:** Visual bars comparing quantities across categories or timeline years.
- **Pie Charts:** Circular graphs showing fractional shares out of a total 360° or 100%.
- **Line Graphs:** Trends over continuous intervals, usually time periods.

2. Solved Examples (Table & Graph Based)

Here are 12 fully solved examples based on standard data representations (Tables, Pie Charts, and Bar Graphs).

Example 1: Basic Table Lookup

Study the following table showing marks of a student in 3 subjects out of 100:

Subject	Maths	Science	English
Marks	78	85	90

Find the total marks and overall percentage obtained by the student.

Solution:

Total Marks = $78 + 85 + 90 = 253$.

Max Total Marks = 300.

Percentage = $(253 / 300) \times 100 = 84.33\%$.

Example 2: Percentage Increase

The sales of a company in 2024 were \$45,000 and in 2025 were \$54,000. Find the percentage increase in sales.

Solution:

Increase in sales = $54,000 - 45,000 = 9,000$.

Percentage Increase = $(9,000 / 45,000) \times 100 = (1 / 5) \times 100 = 20\%$.

Example 3: Ratio Calculation

In a company department, the number of Male employees is 350 and Female employees is 210. Find the ratio of Male to Female employees in simplest form.

Solution:

Ratio = Male : Female = 350 : 210.

Dividing both by their HCF (70):

$350 / 70 = 5$ and $210 / 70 = 3$.

Simplified Ratio = 5 : 3.

Example 4: Average Production

A factory manufactured cars over 4 months: Jan (120), Feb (150), Mar (180), Apr (110). Find the average monthly production.

Solution:

Total production = $120 + 150 + 180 + 110 = 560$.

Number of months = 4.

Average = $560 / 4 = 140$ cars/month.

Example 5: Table Data Analysis

Consider the production of Company A and B (in thousands):

Company	2022	2023	2024
Company A	40	55	60
Company B	30	45	50

What is the total production of Company A across all 3 years?

Solution:

Total production for Company A = $40 + 55 + 60 = 155$ (thousand units).

Example 6: Difference Between Totals

Using the table in Example 5, find the difference between the total production of Company A and Company B over the 3-year period.

Solution:

Company A total = 155.

Company B total = $30 + 45 + 50 = 125$.

Difference = $155 - 125 = 30$ (thousand units).

Example 7: Pie Chart Degree Distribution

In a pie chart representing a total budget of \$360,000, the sector angle for "Marketing" is 54° . Find the amount spent on Marketing.

Solution:

Total angles in a circle = 360° correspond to \$360,000.

Value for $1^\circ = \$360,000 / 360 = \$1,000$.

Marketing Amount = $54 \times 1000 = \$54,000$.

Example 8: Pie Chart Percentage Share

In a pie chart, expenditure on Food constitutes 35% of total monthly expenses. If total monthly expenses are \$20,000, how much is spent on Food?

Solution:

Food Expense = 35% of \$20,000.

$(35 / 100) \times 20,000 = 35 \times 200 = \$7,000$.

Example 9: Percentage Comparison

If Company X's revenue is \$80 million and Company Y's revenue is \$60 million, Company X's revenue is what percentage more than Company Y's?

Solution:

Difference = $80 - 60 = 20$ million.

Percentage more than Y = $(20 / 60) \times 100 = (1 / 3) \times 100 = 33.33\%$.

Example 10: Growth Rate Between Two Years

The population of a town was 20,000 in 2020 and grew to 23,000 in 2022. Find the annual growth rate (simple average per year).

Solution:

Total increase = $23,000 - 20,000 = 3,000$ over 2 years.

Annual increase = $3,000 / 2 = 1,500$ per year.

Percentage annual growth = $(1,500 / 20,000) \times 100 = 7.5\%$.

Example 11: Weighted Average Calculation

Class X has 30 students with an average score of 60, and Class Y has 20 students with an average score of 80. Find the combined average score of all students.

Solution:

Total marks for Class X = $30 \times 60 = 1800$.

Total marks for Class Y = $20 \times 80 = 1600$.

Combined Total Marks = $1800 + 1600 = 3400$.

Total Students = $30 + 20 = 50$.

Combined Average = $3400 / 50 = 68$.

Example 12: Ratio of Averages

The total export of country P over 3 years is 450 tons, and country Q over 3 years is 600 tons. Find the ratio of their average exports.

Solution:

Average of P = $450 / 3 = 150$.

Average of Q = $600 / 3 = 200$.

Ratio = $150 : 200 = 3 : 4$.

3. Practice Questions

Solve these 25 multiple-choice questions to test your proficiency in Data Interpretation. An answer key is provided on the final page.

Q1. If the total expenditure of a company is \$500,000 and the sector angle for 'R&D' is 72° , what is the expenditure on R&D?

- | | |
|--------------|--------------|
| A) \$100,000 | B) \$72,000 |
| C) \$50,000 | D) \$120,000 |

Q2. The ratio of boys to girls in a school is 5 : 3. If there are 400 girls, how many boys are there?

- | | |
|--------|--------|
| A) 667 | B) 600 |
| C) 750 | D) 500 |

Q3. A company's profit increased from \$20 million to \$25 million. What is the percentage growth?

- | | |
|--------|--------|
| A) 20% | B) 25% |
| C) 15% | D) 30% |

Q4. Average of five numbers is 40. If one number is excluded, the average becomes 35. What is the excluded number?

- | | |
|-------|-------|
| A) 55 | B) 60 |
| C) 45 | D) 50 |

Q5. If A is 25% more than B, then B is what percentage less than A?

- | | |
|--------|--------|
| A) 25% | B) 20% |
| C) 15% | D) 10% |

Q6. In a pie chart representing 100%, a category taking 45 degrees corresponds to what percentage?

- | | |
|----------|--------|
| A) 12.5% | B) 15% |
| C) 10% | D) 20% |

Q7. If sales in Jan, Feb, Mar are 100, 150, and 200 respectively, find the percentage increase from Jan to Mar.

- | | |
|---------|---------|
| A) 50% | B) 100% |
| C) 200% | D) 150% |

Q8. The population of a city rises from 150,000 to 165,000 in one year. Find the growth rate.

- | | |
|--------|----------|
| A) 5% | B) 10% |
| C) 15% | D) 12.5% |

Q9. If 40% of a number is 120, what is 60% of that number?

- | | |
|--------|--------|
| A) 150 | B) 180 |
| C) 200 | D) 240 |

Q10. The average of 4 numbers is 18. If each number is multiplied by 2, what is the new average?

- | | |
|-------|-------|
| A) 18 | B) 36 |
| C) 72 | D) 9 |

Q11. If the cost price is \$200 and selling price is \$250, what is the profit percentage?

- | | |
|--------|--------|
| A) 20% | B) 25% |
| C) 15% | D) 30% |

Q12. Out of 500 students, 350 passed an exam. What is the failure percentage?

- | | |
|--------|--------|
| A) 30% | B) 25% |
| C) 35% | D) 20% |

Q13. If $x : y = 3 : 4$ and $y : z = 8 : 9$, find $x : z$.

- | | |
|----------|----------|
| A) 1 : 2 | B) 2 : 3 |
| C) 3 : 4 | D) 1 : 3 |

Q14. A car travels 120 km in 2 hours and another 180 km in 3 hours. What is the average speed?

- | | |
|------------|------------|
| A) 55 km/h | B) 60 km/h |
| C) 65 km/h | D) 50 km/h |

Q15. If 15 workers complete a task in 8 days, how many days will 20 workers take?

- | | |
|-----------|-----------|
| A) 5 days | B) 6 days |
| C) 7 days | D) 4 days |

Q16. The difference between 60% and 40% of a number is 50. What is the number?

- | | |
|--------|--------|
| A) 200 | B) 250 |
| C) 300 | D) 150 |

Q17. In a survey of 1000 people, 650 liked coffee and 400 liked tea. If 200 liked both, how many liked neither?

- | | |
|--------|--------|
| A) 100 | B) 150 |
| C) 50 | D) 200 |

Q18. If the simple interest on a sum for 2 years at 10% per annum is \$400, what is the principal amount?

- | | |
|------------|------------|
| A) \$1,000 | B) \$2,000 |
| C) \$1,500 | D) \$2,500 |

Q19. If a bar graph shows values 40, 60, 80, 50 across 4 years, what is the total sum?

- | | |
|--------|--------|
| A) 220 | B) 230 |
| C) 240 | D) 250 |

Q20. What is the angle in a pie chart for a component representing 25% of the total?

- | | |
|---------|--------|
| A) 60° | B) 90° |
| C) 120° | D) 45° |

Q21. If the ratio of incomes of P and Q is 3 : 4 and their expenditures are in ratio 2 : 3, and both save \$200, find P's income.

- | | |
|------------|------------|
| A) \$1,200 | B) \$1,600 |
| C) \$800 | D) \$1,000 |

Q22. If 32% of a number is subtracted from itself, it reduces to 408. What is the number?

- | | |
|--------|--------|
| A) 550 | B) 600 |
| C) 650 | D) 500 |

Q23. The average weight of 10 men increases by 1.5 kg when a new man replaces one weighing 60 kg. Find the weight of the new man.

A) 70 kg

B) 75 kg

C) 80 kg

D) 65 kg

Q24. If the radius of a circle is increased by 10%, by what percentage does its area increase?

A) 10%

B) 21%

C) 20%

D) 12.1%

Q25. If a train runs at 72 km/h, what is its speed in meters per second (m/s)?

A) 15 m/s

B) 20 m/s

C) 25 m/s

D) 18 m/s

Answer Key

Q1: A	Q2: B	Q3: B	Q4: B	Q5: B
Q6: A	Q7: B	Q8: B	Q9: B	Q10: B
Q11: B	Q12: A	Q13: B	Q14: B	Q15: B
Q16: B	Q17: B	Q18: B	Q19: B	Q20: B
Q21: A	Q22: B	Q23: B	Q24: B	Q25: B